

TALI Modular Thoracic Auxiliary Limb Exosuit: Official Design Master Plan

A Black-and-White APA-Style Technical Design Paper for a Non-Surgical, Interchangeable
Auxiliary-Limb Platform

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Project TALI

Design edition: Adult Alpha Prototype and Integrated Model Architecture

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Document status: Official project design specification. This document defines the project architecture, goals, modular product family, first adult working prototype, and engineering master plan. It is not a research update, preliminary summary, or literature review. Evidence is cited only to support design decisions, component selection, interface control, and standards alignment.

Abstract

This design paper defines the TALI Modular Thoracic Auxiliary Limb Exosuit as a non-surgical wearable robotics platform derived from the anatomical logic of Thoracic Auxiliary Limb Integration while replacing surgical integration with an external thoraco-pelvic exoframe. The project consists of one shared wearable base, designated TALI-Core, and two distinct interchangeable mission models: TALI-H, a hand-arm auxiliary manipulation model, and TALI-W, a deployable wing-arm model. The models are not merged into a single mixed morphology; they share power, control, safety, sensing, docking, sizing, and training infrastructure while retaining separate mechanical modules, work envelopes, end-effectors, control profiles, and validation pathways. TALI-H is intended for quadrimanual coordination, tool handling, machinery interface, bracing, clamping, and dexterous or semi-dexterous manipulation. TALI-W is intended for foldable, spar-like posterior appendages that provide reach, stowage, bracing, stabilisation, and future tethered aerodynamic investigation without treating wearable flight as a first-generation objective. The first adult working prototype is defined as an externally powered, thoraco-pelvic, single-module-capable platform with a universal thoracic dock, low distal mass, torso-mounted actuation, mechanical resting locks, modular end-effectors, body-signal control, shared autonomy, and adult fit commissioning. Developmental candidate training is specified as a longitudinal curriculum beginning with simulation, extra-thumb or extra-finger training, passive modules, low-force wearable modules, and adult high-authority operation. This paper presents the official project architecture, design requirements, module separation, hardware concept, control stack, software architecture, first prototype configuration, component candidates, and implementation master plan.

Keywords: thoracic auxiliary limb integration, supernumerary robotic limbs, exosuit, wearable robotics, quadrimanual coordination, robotic hands, wing-arms, non-invasive neural interface, human-machine interface, modular exoframe

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Project Definition and Design Doctrine

The TALI Modular Thoracic Auxiliary Limb Exosuit is an external, non-surgical wearable robotics system that uses the thoracic anatomical logic of a second limb-girdle concept without requiring implantation, surgical nerve routing, biological tissue integration, or permanent body modification. The originating TALI concept describes Thoracic Auxiliary Limb Integration as a conceptual bio-anatomical design for a secondary pair of arms emerging from a reinforced upper-to-mid thoracic structure integrated with the rib cage and spine (IUSmusic, 2026). In the official exosuit project, that anatomical premise is translated into a removable thoraco-pelvic exoframe, a universal thoracic docking system, and interchangeable auxiliary modules rather than an implanted skeletal or neural structure.

The design doctrine is modular integration without morphological mixing. The platform contains two distinct models: TALI-H, the auxiliary hand-arm manipulation model, and TALI-W, the wing-arm deployment and stabilisation model. The two models must be able to use the same core suit, the same dock standard, the same control computer, the same candidate profile database, the same power and safety bus, and the same training environment. They must not be mechanically or conceptually collapsed into one hybrid limb. TALI-H is an arm-and-hand system. TALI-W is a foldable wing-arm system. Integration occurs at the platform, dock, software, control, and training layers.

The project is intended to produce a real working adult prototype before any broad protective gating, paediatric deployment gate, industrial certification gate, or clinical-style gate constrains the concept. This does not remove engineering safeguards from the artifact itself. Instead, safeguards are built into the design as operating requirements: mechanical stops, low-force adult commissioning modes, offboard power, emergency release, joint locking, collision-aware software, and sensor feedback. The project therefore distinguishes between research gating and engineering design controls. Research gating is not used as a reason to reduce the design prior to the first adult working prototype. Engineering controls are part of the prototype because the object is a human-worn robot.

The official project scope is non-medical, non-surgical, adult-prototype-first wearable augmentation. Surgical TALI remains a separate anatomical concept and is not claimed as the near-term physical system. TALI-Core, TALI-H, and TALI-W are external devices. If later applications include surgery-adjacent tool support, rehabilitation, clinical training, public safety, military, or industrial deployment, those domains require separate downstream compliance tracks. This document defines the common design foundation that can later support those tracks.

Terminology

Term	Definition in this project
TALI	Thoracic Auxiliary Limb Integration as the originating anatomical concept and project family.
TALI-Core	The shared thoraco-pelvic exoframe, dock, power, sensing, control, and training platform.
TALI-H	The hand-arm auxiliary module for manipulation, grasping, machinery interface, and quadrimanual coordination.
TALI-W	The wing-arm auxiliary module for foldable deployment, bracing, stabilisation, and wide-reach support.
Universal Thoracic Dock (UTD)	The mechanical, electrical, data, fluid, and safety interface that connects TALI-Core to either TALI-H or TALI-W.

Term	Definition in this project
Candidate	A trained wearer of the system. The term includes adult prototype operators and later developmental trainees.
Adult Alpha Prototype	The first working wearable system intended for competent adult human commissioning under controlled engineering conditions.

Table 1. Core terminology used throughout the official design.

Project Goals

The project goal is to create a non-surgical thoracic auxiliary exosuit system that enables trained candidates to operate additional mechanical appendages in parallel with their biological arms. The system is not designed as a costume, passive backpack, or aesthetic prop. It is a wearable robot architecture for studying and producing usable quadrimanual coordination, tool assistance, machinery interaction, and load stabilisation through long-term training and modular hardware.

The primary objective is an adult working prototype that proves the following: a human adult can wear a thoraco-pelvic TALI-Core; the core can accept interchangeable TALI-H and TALI-W modules; the attached module can deploy, rest, lock, brace, grasp, or manipulate under shared human-machine control; the candidate can command the system through body-signal interfaces without surgical implantation; and the system can operate using hybrid external energy and body-derived control rather than relying on a heavy all-battery suit.

Primary Design Goals

- Create a single shared wearable core capable of accepting two distinct auxiliary-limb models without redesigning the harness, control computer, or power architecture.
- Develop TALI-H as a practical auxiliary hand-arm module for tool use, gripping, clamping, industrial manipulation, machinery interface, and quadrimanual training.
- Develop TALI-W as a foldable wing-arm module for reach extension, bracing, stabilisation, stowage efficiency, deployable support, and future tethered aerodynamic experimentation.
- Use an external thoraco-pelvic frame that preserves thoracic anatomical placement while transferring load to the pelvis and posterior trunk rather than to the rib cage, axilla, clavicle, or neck.
- Use modular end-effectors rather than forcing a single flesh-like hand to perform all tasks.
- Use non-surgical control interfaces: sEMG, foot pressure, toe input, IMUs, gaze/head targeting, respiration, tongue/mouth switching, voice mode selection, and optional EEG/EEG-fNIRS only as supervisory interfaces.
- Use hybrid quasi-passive power: spring or elastic support, mechanical locking, external electrical supply, optional pneumatic/vacuum services, and a small onboard battery for compute, sensing, and safe retraction.
- Create a longitudinal training pathway through simulation, extra-thumb/extra-finger devices, passive modules, low-force modules, and finally adult high-authority wearable modules.

Non-Goals of the First Adult Prototype

The first adult prototype is not a surgical implant, a flying suit, a clinical medical device, a paediatric load-bearing exoskeleton, a fully autonomous industrial robot, a direct brain-only robotic arm, or a full humanoid replication of natural shoulders and hands. These exclusions define the first build boundary, not the long-term imagination of the project.

System-Level Architecture

The TALI architecture is a platform-and-module system. TALI-Core provides the human-worn structure, power distribution, sensing, safety interlocks, compute, software, and training identity. TALI-H and TALI-W are mission modules. The modules attach to the same universal thoracic docks and announce their identity to the software. The software loads the appropriate kinematic model, control profile, limit set, end-effector library, and training mode for the attached module.

TALI system family

```
Human candidate
|-- soft base garment and pressure interface
|-- TALI-Core thoraco-pelvic exoframe
    |-- posterior spine shell
    |-- pelvic/iliac load-transfer belt
    |-- sternum stabiliser
    |-- power/data/safety bus
    |-- left and right Universal Thoracic Docks
        |-- TALI-H module: auxiliary hand-arm
        |-- TALI-W module: auxiliary wing-arm
        |-- passive training module
    |-- TALI-OS / ROS 2 control layer
    |-- candidate profile and training database
```

Figure 1. Text-only black-and-white system decomposition of the TALI platform family.

The operating principle is that the candidate gives high-level intent, the wearable collects body-derived command signals, the shared autonomy layer resolves movement primitives, and the module executes force-limited motion. The robot does not require the candidate to teleoperate every joint. Instead, the control grammar uses commands such as deploy, fold, rest, lock, open, close, hold, brace, stabilise, release, follow target, and return. The specific response depends on whether the attached module is TALI-H or TALI-W.

Layer	Shared by both models	Model-specific difference
Human interface	Garment, pressure mapping, sizing, don/doff, emergency release	Different load posture and user cues for hand-arm versus wing-arm modules
Frame	Thoraco-pelvic load path and posterior dock geometry	TALI-H uses shorter close-body manipulation geometry; TALI-W uses longer foldable spar geometry
Power	External power, local battery for control, joint brakes, sensor bus	Different peak power and locking modes per module
Control	sEMG, foot, IMU, gaze, optional EEG; shared autonomy	TALI-H emphasizes grasp primitives; TALI-W emphasizes deployment and bracing primitives
End-effectors	Common quick-change wrist/tool protocol	Hands and grippers for TALI-H; brace pads, spars, clamps, hooks, or tool supports for TALI-W
Software	TALI-OS, candidate profile, module identification, logs	Separate kinematic models, workspace limits, user mappings, and task libraries

Table 2. Integration model for keeping TALI-H and TALI-W distinct while using one core platform.

Model A: TALI-H Hand-Arm Auxiliary Module

TALI-H is the hand-arm auxiliary model. Its primary design purpose is manipulation. It is the correct module for machinery interaction, tool use, object transfer, clamping, holding, stabilisation, surgery-adjacent instrument support, laboratory manipulation, and quadrimanual training. It should be interpreted as an external auxiliary arm system, not as a biological replication of natural arms.

The recommended TALI-H morphology is asymmetric when used as a two-module pair. One side should be optimized as a manipulator with an adaptive hand or anthropomorphic underactuated hand. The other side should be optimized as a support arm with a rugged two-finger gripper, three-jaw gripper, hook, clamp, jamming gripper, vacuum pad, or tool cradle. This avoids the engineering error of building two complex humanoid hands when many real tasks require one manipulator and one stabiliser.

Mechanical Architecture of TALI-H

A TALI-H module should use three powered gross-positioning degrees of freedom for the first adult prototype: thoracic-root yaw/sweep, elevation/pitch, and elbow/fold or reach extension. A passive, lockable, or low-power wrist is sufficient at first. The hand or gripper is the primary distal powered subsystem. Distal motors should be minimized because the wrist and hand have the highest leverage penalty on perceived mass and inertial disturbance.

The module should use a short, strong proximal link and a lighter distal link. All heavy motors should be torso-mounted or root-mounted and transmit torque through cables, belts, or compact gearboxes close to the body. Where a powered joint may contact the wearer or an external object, the joint should include compliance, torque limitation, current monitoring, speed limitation, and mechanical hard stops.

Functional Modes of TALI-H

Mode	Description	Primary inputs	Primary output
Rest	Arm folded close to posterior torso or flank with joint brakes engaged.	Mode switch, inactivity timer	Low-power locked posture
Deploy	Arm unfolds to predefined safe workspace outside head and neck zone.	Foot command, gaze target, EMG confirmation	Position-ready state
Hold	End-effector grips or supports selected object while arm compensates for body motion.	Gaze/target selection, grip command	Stable object retention
Brace	Arm forms a static support contact against tool, surface, rail, or fixture.	Foot command, pressure feedback	Structural support and force distribution
Manipulate	Arm performs slow force-limited motion under shared autonomy.	sEMG, foot pressure, head/gaze	Controlled displacement, grasp, release
Tool assist	Arm holds tool body, camera, light, cable, instrument, or handle while biological hands perform fine work.	Task primitive selection	Third/fourth hand function
Emergency retract	Arm stops, releases or holds according to hazard class, and folds to safe posture.	E-stop, signal loss, fault	Safe de-energized or locked state

Table 3. TALI-H functional modes.

TALI-H End-Effector Classes

TALI-H requires a modular end-effector ecosystem. Human-like hands are useful for anthropomorphic tasks and embodiment training, but industrial and heavy-equipment interaction requires grippers, clamps, and tool-specific endpoints. The qb SoftHand Industry is a candidate adaptive anthropomorphic hand because it is described by the manufacturer as a tendon-driven industrial hand with five human-like fingers, 19 anthropomorphic degrees of freedom controlled by one synergy motor, a 2 kg power-grasp payload, 0.99 kg weight, IP65 resistance, and ROS/ROS2 compatibility (qbrobotics, n.d.). The Robotiq 2F-85 is a candidate support gripper because it is specified with an 85 mm stroke, 5 kg payload, approximately 1 kg gripper weight, 20 to 235 N grip force, and 24 V electrical operation (Robotiq, n.d.).

The official design position is that the first adult prototype should use a rugged gripper before a complex dexterous hand, then add an adaptive hand after gross control, load transfer, and safety performance are proven in adults. The manipulator side may use an adaptive hand. The support side should initially use a robust gripper or clamp.

Model B: TALI-W Wing-Arm Auxiliary Module

TALI-W is the wing-arm auxiliary model. It is a deployable, foldable, spar-like posterior appendage system. It is not the same as TALI-H and must not be treated as a humanoid hand-arm with decorative wings. TALI-W uses wing-like morphology for folding, stowage, bracing, broad workspace, and load distribution. Its first purpose is not untethered flight. Its first purpose is reach extension, stabilisation, bracing, tool support, and deployable structural assistance.

The TALI-W model should inherit the broad thoracic origin suggested by the TALI anatomical concept while externalising the load path through the thoraco-pelvic frame. The wing-arm root is a high-load proximal joint mounted posterolaterally to the thoracic dock. A short strong proximal spar emerges from the root, followed by one or two lightweight foldable distal spars that can stow along the back or flanks. Endpoints may include brace pads, hooks, clamping elements, tool holders, jamming grippers, camera/light mounts, or specialised support effectors.

Mechanical Architecture of TALI-W

TALI-W should use folding geometry rather than a humanoid elbow and wrist chain. The root must accept high moment loads and must transfer them through the posterior backplate into the pelvic belt. Distal elements should use carbon-fibre tubes, thin composite shells, or aluminium lattice structures to reduce inertial burden. The wing-arm should lock in several discrete postures: fully folded, partially deployed, lateral brace, overhead support, rear support, and forward tool-support posture.

TALI-W wing-arm concept

```

Thoracic dock
|-- root yaw/elevation joint
    |-- short proximal spar
        |-- fold joint with brake
            |-- lightweight distal spar
                |-- endpoint node
                    |-- brace pad / hook / clamp / gripper / tool holder

```

Figure 2. Text-only wing-arm module decomposition.

Functional Modes of TALI-W

Mode	Description	Primary mechanical requirement
Folded rest	Wing-arm lies along posterior torso or flank with low protrusion.	Compact stowage, lockable joints
Deploy	Spar unfolds to selected workspace without entering face/neck exclusion zone.	Controlled trajectory and spatial limits
Brace	Endpoint presses or locks against surface, rail, floor-contact accessory, fixture, tool body, or workpiece.	Force path through root and pelvis
Stabilise	Wing-arm damps body or tool motion, acting as an external stabilising appendage.	Impedance control and load sensing
Wide reach	Distal spar carries a light tool, sensor, camera, cable, or support device to a larger radius than TALI-H.	Low distal mass and collision monitoring
Absorb load	Wing-arm resists external disturbance or transfers load into frame.	Mechanical brakes, compliant contact, root strength
Retract	Wing-arm folds to a safe low-energy state after task or fault.	Fail-safe spring/bias or powered retraction

Table 4. TALI-W functional modes.

TALI-W and Flight Boundary

TALI-W is a working wing-arm system, not a first-generation human flight system. The term wing-arm refers to deployable morphology and structural function. If future research explores lift, drag, or tethered aerodynamic assistance, it should be a separate downstream module family. The adult prototype must not depend on aerodynamic lift to meet its project objectives. The official first functional target is ground-dominant augmentation: brace, hold, reach, stabilise, and support.

TALI-Core Thoraco-Pelvic Exoframe

TALI-Core is the common wearable base. It must be strong enough to support either TALI-H or TALI-W while remaining comfortable for trained candidates. The frame is thoracic in module placement and control geometry, but thoraco-pelvic in load transfer. Loads generated by auxiliary modules must not terminate in the rib cage. They must flow through a posterior spinal member and pelvic belt, using the trunk as a stabilised interface rather than a load-bearing biological socket.

Core Structure

TALI-Core consists of five physical layers: a soft base garment, a pressure-distributing textile interface, a rigid posterior spine shell, a pelvic load-transfer belt, and left/right universal thoracic docks. The soft garment provides comfort, moisture management, sensor routing, and pressure sensor locations. The textile interface distributes contact pressure and prevents hard axillary, clavicular, or rib loading. The posterior spine shell is a semi-rigid structural member spanning upper thorax to pelvis. The pelvic belt transfers vertical and torsional loads to the iliac region. The docks provide module attachment and release.

Component	Design requirement	Rationale
Soft garment	Breathable, washable, replaceable, sensor-compatible, adjustable	Maintains hygiene, comfort, and repeatable sensor placement
Posterior spine shell	Carbon composite or aluminium-composite; contoured; service-channel ready	Carries module forces from thoracic root to pelvis

Component	Design requirement	Rationale
Pelvic/iliac belt	Rigid enough for load transfer; adjustable; padded at iliac crest	Avoids suspending loads from ribs and shoulders
Sternum stabiliser	Limits rotation without restricting breathing	Prevents frame twist and dock migration
Scapular cut-outs	Clear biological shoulder blades and natural arm motion	Reduces collision and skin pressure
Dock mounts	Left/right symmetric; keyed; mechanically redundant; sensorized	Allows module interchangeability and load monitoring
Emergency doffing	Single-action release path accessible to wearer and supervisor	Supports rapid removal after fault or discomfort

Table 5. TALI-Core physical components.

Sizing System

The platform must support adult alpha operation first, then later staged developmental versions. Adult sizing should be modular across torso length, shoulder breadth, chest circumference, waist circumference, iliac crest position, and scapular clearance. Size adjustment should not rely on flexible straps alone. Structural members must have repeatable adjustment positions, indexed fasteners, and calibration metadata so that the software knows the frame configuration used during operation.

For developmental training, the project should not simply scale down the adult load-bearing system. The official training ecosystem uses non-load-bearing simulation, extra-digit devices, passive modules, and low-force rigs before any juvenile thoracic load-bearing system. This preserves the goal of early motor skill development while keeping the high-force exosuit as an adult prototype object until later evidence and design maturity justify broader age ranges.

Universal Thoracic Dock

The Universal Thoracic Dock (UTD) is the most important integration decision in the project. It is the interface that makes two distinct models usable on one platform. Each TALI-Core should provide left and right UTD locations on the posterior thoracic frame. Each dock must support TALI-H, TALI-W, passive training modules, dummy ballast modules, and test adapters. The dock must prevent attachment errors and must identify the module before power is enabled.

Dock subsystem	Required function
Mechanical mount	Carries bending, torsion, axial, and shear loads; uses keyed geometry to prevent incorrect orientation.
Redundant lock	Provides primary mechanical latch and secondary lock indicator before activation.
Power connector	Supplies regulated module power with pre-charge and fault detection.
Data connector	Provides CAN-FD, EtherCAT, Ethernet, or equivalent deterministic module communication.
Safety interlock	Reports latch status, module identity, temperature, electrical fault, and emergency release status.
Fluid/service line	Optional pneumatic, vacuum, or cooling interface for jamming grippers, suction tools, or pneumatic brakes.
Load sensor	Measures dock moment/load class to protect the wearer and frame.
Module ID	Reads embedded module identity, serial number, firmware compatibility, calibration file, and maximum allowed operating profile.

Dock subsystem	Required function
Manual release	Allows controlled mechanical detachment when module is unloaded and powered down.

Table 6. Universal Thoracic Dock requirements.

Modules are swappable but not hot-swappable under load. A module change requires return to rest, joint locking, power disablement, latch release, module removal, new module attachment, identification, software profile loading, calibration check, and safe activation. The system should physically prevent unlatching while a module reports load above a defined safe detachment threshold.

Skeletal System and Human Interface

The skeletal system consists of the external structural elements that connect module forces to the candidate. It is not a biological skeleton, but it is designed around the anatomical reality of the thorax, scapulae, spine, pelvis, breathing motion, and natural arm range of motion. The system must remain comfortable enough for repeated training and strong enough for meaningful adult prototype operation.

Load Path

The skeletal load path is defined as follows: module endpoint load to distal link; distal link to root joint; root joint to UTD; UTD to posterior thoracic backplate; backplate to posterior spine shell; spine shell to pelvic/iliac belt; pelvic belt to wearer stance or external support. The chest and ribs are stabilising contact zones, not primary load-bearing elements. The shoulder straps and textile vest are alignment structures, not suspension straps for module torque.

Endpoint load -> module links -> thoracic dock -> posterior spine shell
-> pelvic load-transfer belt -> wearer stance or external support

Forbidden primary load paths:
Endpoint load -> axilla
Endpoint load -> clavicle
Endpoint load -> compressive chest ring
Endpoint load -> cervical/neck straps

Figure 3. Permitted and forbidden load paths.

Human Contact Requirements

The interface must protect breathing, scapular movement, skin integrity, circulation, and nerve pathways. The design shall not include hard axillary hooks, unpadded clavicular compression, circumferential rigid rib compression, pressure concentration over the spinous processes, or a load-bearing neck yoke. Sensorized pads should monitor pressure at the sternum, lateral ribs, iliac crest, scapular border, lumbar pad, and posterior thoracic shell.

Candidate comfort is a functional requirement, not an accessory. A device that can only be worn briefly under discomfort will not allow longitudinal motor learning. The frame must support repeated donning, consistent fit, pressure telemetry, replaceable contact materials, and a defined fitting procedure.

Power and Actuation Architecture

The official TALI power doctrine is hybrid quasi-passive. The exosuit should not be a fully battery-powered heavy suit. It should use external energy for high-load adult prototype operation, passive or

elastic support for posture and gravity compensation, mechanical locks for rest states, and a small onboard battery only for compute, sensors, hand electronics, and safe retraction or controlled shutdown.

Power Modes

Power mode	Application	Design intent
External electrical tether	Adult prototype lab, workshop, controlled test stand	Provides high energy without requiring the candidate to carry large batteries
Onboard control battery	Compute, sensing, communication, low-energy release/retract	Maintains controller and safety operation under tether fault
Passive elastic support	Gravity compensation, partial arm weight relief, wing-arm deployment support	Reduces continuous motor torque and fatigue
Mechanical brake/lock	Static holding, rest, bracing, power-off posture	Maintains position without continuous current
Optional pneumatic/vacuum	Jamming gripper, suction pad, pneumatic brake, soft actuator experiment	Expands tool ecosystem with offboard service line
External structural brace	High reaction force tasks	Transfers heavy task forces to workstation, rail, trolley, or ground contact instead of wearer alone

Table 7. TALI power modes.

Actuation Strategy

The first adult prototype should prioritise torso-mounted or root-mounted actuators. Heavy motors should not be located at the distal wrist or hand unless the selected end-effector inherently includes a motor. Gross movement should be generated by actuators close to the spine shell or pelvic pack and transmitted by Bowden cable, belt, tendon, or compact root gear transmission. Joint design should include compliance where unexpected contact is plausible.

Holding is a structural task. The device should not consume high continuous motor current merely to hold a static posture. The system should use spring balancing, ratchet or clutch locks, electromagnetic brakes, worm or harmonic gear self-holding where appropriate, and software-controlled low-power hold modes. TALI-W in particular requires discrete mechanical deployment locks because long spars create large moments at the root.

End-Effector and Hand System

The end-effector system is modular by design. It includes adaptive hands, industrial grippers, tool holders, clamps, hooks, brace pads, vacuum pads, jamming grippers, and training endpoints. The project should not require the auxiliary limb to look like flesh. Functional grip, reliability, contact safety, force feedback, cleaning, tool compatibility, and low distal mass are more important than cosmetic realism.

End-Effector Families

Family	Primary use	Recommended first use
Soft/adaptive hand	Irregular object grasping, anthropomorphic skill training, safe contact	TALI-H manipulator side after gripper prototype
Two-finger parallel gripper	Machinery interface, tool holding, parts handling, clamps	TALI-H support side and first adult prototype
Three-jaw gripper	Cylindrical tools, handles, rods, fixtures	Industrial TALI-H toolset

Family	Primary use	Recommended first use
Jamming gripper	Irregular object support with low wrist complexity	TALI-W or support-arm variant
Hook/cradle	Cable, hose, bucket, tray, tool, rail engagement	TALI-W endpoint library
Vacuum pad	Panels, sheets, clean surfaces	Workshop/lab accessory
Brace pad	Structural contact against body, machine, wall, vehicle, bench, or floor accessory	TALI-W core endpoint
Instrument holder	Camera, light, scope, lab instrument, surgery-adjacent tool support	Specialist TALI-H/TALI-W accessory
Training endpoint	Foam hand, passive spar, low-force thumb or finger	Candidate curriculum

Table 8. End-effector families.

Quick-Change Interface

Each terminal node should use a compact quick-change interface. OnRobot specifies a Quick Changer with approximately 200 g weight, an ISO-9409-1 flange, redundant locking, no screwing required for attachment, and payload capacity up to 20 kg (OnRobot, n.d.). Equivalent interfaces may be used if they meet the same repeatability, locking, load, and sensor requirements. The official design should preserve mechanical interchangeability and avoid hardwiring a single end-effector into the module.

The hand system is therefore a library, not a single component. The adult prototype should ship with at least four end-effector classes: rugged gripper, soft/adaptive training hand, brace pad or hook, and dummy safe endpoint for calibration.

Control Interfaces and Non-Surgical Brain-Signal Layer

The TALI control system is non-surgical. It must not require implanted electrodes, nerve transfer surgery, or permanent body modification. The project uses a layered command system: body-derived signals for primary control; shared autonomy for trajectory and grasp details; and optional non-invasive brain signals for supervisory mode selection, confirmation, workload estimation, or accessibility. Brain-only control is not the first adult prototype command architecture.

Primary Control Stack

Input channel	Use in TALI	Notes
sEMG	Grip intent, stiffness modulation, mode confirmation, limb selection	Delsys describes EMG as measuring muscle electrical activity and as applicable to robotics and human-machine interaction (Delsys, n.d.).
Foot pressure / toe input	Continuous low-dimensional command, mode selection, extra thumb/finger training transfer	Allows biological hands to remain available.
IMUs	Torso, limb, biological arm, foot, and head motion tracking	Supports body-motion compensation and candidate posture model.
Gaze/head tracking	Target selection and workspace attention cue	Best for selecting what the robot should act on, not for direct high-DOF motion alone.
Respiration/diaphragm	Mode switching or proportional low-rate command	Useful where feet or hands must remain occupied.

Input channel	Use in TALI	Notes
Tongue/mouth interface	Discrete mode control and accessibility	Useful for candidates requiring hands-free and foot-free command options.
Voice	Non-time-critical mode selection and supervisor commands	Not suitable as a sole safety-critical real-time controller.
EEG / EEG-fNIRS	Supervisory intent, workload, confirmation, accessibility layer	Not used as the primary low-level controller for heavy modules.

Table 9. Non-surgical control input channels.

Delsys Trigno is an example research-grade sEMG ecosystem suitable for early integration because it is described as a wireless EMG system for real-world data capture, human-machine interaction, robotics, and API-enabled development (Delsys, n.d.). Wearable Sensing DSI-24 is an example dry EEG system for the optional BCI layer because it provides wireless dry EEG, less-than-three-minute setup, full head coverage with 19 head electrodes, auxiliary inputs, trigger synchronization, and software/API access (Wearable Sensing, n.d.). OpenBCI Cyton is a lower-cost open documentation ecosystem suitable for exploratory EEG/EMG acquisition and early software experimentation (OpenBCI, n.d.).

Control Philosophy

The candidate should command goals, modes, and primitives. The robot should handle joint-level details. This is especially important for TALI-W, where the candidate should not need to consciously steer every spar joint. It is also important for TALI-H, where grasp success depends on contact, slip, alignment, and force regulation. The system should therefore expose a compact command vocabulary rather than continuous high-dimensional teleoperation.

Candidate command	TALI-H interpretation	TALI-W interpretation
Rest	Fold arm close to torso and lock.	Fold wing spar along back/flank and lock.
Deploy	Move arm to safe ready posture near selected workspace.	Unfold spar to deployment posture.
Hold	Grip selected object and maintain force.	Brace or support selected object/surface.
Stabilise	Damp tool or object motion through end-effector.	Create broad structural support against external contact.
Release	Open gripper or relax contact under safe trajectory.	Unload brace and retract endpoint.
Follow	Track gaze, biological hand, tool, or selected reference.	Maintain relative position or support envelope.
Lock	Engage mechanical hold posture.	Engage spar/root/fold locks.

Table 10. Shared command vocabulary with model-specific behavior.

Software Architecture and Shared Autonomy

The software architecture should use ROS 2 and `ros2_control` or equivalent middleware for modular robotic control. ROS 2 documentation identifies active distributions and provides packages for topics, services, actions, parameters, transforms, URDF, simulation, launch, testing, and middleware configuration (Open Robotics, 2026). The `ros2_control` project provides a framework for robot hardware interfaces and controllers (ROS Controls, 2026). The TALI software layer is designated TALI-OS. TALI-OS is not a consumer operating system; it is the project control and integration architecture.

TALI-OS architecture

Safety kernel

- |-- emergency stop and fault states
- |-- joint, speed, force, and workspace limits
- |-- dock latch and module identity checks

Hardware abstraction

- |-- motors, encoders, brakes, grippers, hand electronics
- |-- force/torque, pressure, IMU, EMG, EEG, cameras

Perception and state estimation

- |-- candidate posture
- |-- module kinematics
- |-- object target and contact state
- |-- harness pressure and load path

Shared autonomy

- |-- deployment planner
- |-- grasp primitive manager
- |-- bracing primitive manager
- |-- collision-aware motion
- |-- body-motion compensation

Candidate profile

- |-- calibration
- |-- preferred input mappings
- |-- motor-learning progression
- |-- fatigue and performance metrics

Training environment

- |-- simulator
- |-- curriculum tasks
- |-- telemetry dashboard
- |-- module-specific skill tree

Figure 4. TALI-OS functional architecture.

Autonomy Boundaries

Shared autonomy should never eliminate candidate authority. The candidate selects intent, task, target, or mode. The robot resolves low-level motion within strict constraints. The autonomy layer may align a gripper, compensate for torso movement, maintain brace force, damp vibration, avoid forbidden body zones, or execute a calibrated grasp primitive. The autonomy layer may not unexpectedly initiate high-force motion near the body, face, neck, or bystanders.

Candidate Development and Motor-Skill Curriculum

The project goal includes early skill introduction and long-term development of candidates. The official design separates early learning from high-force hardware. Candidates should develop supernumerary control through stages: virtual limbs, extra-thumb/extra-finger devices, passive training modules, low-force wearable modules, and adult high-authority operation. This preserves the project ambition to develop motor skills over years while maintaining the adult working prototype as the first load-bearing human exosuit milestone.

Curriculum Stages

Stage	Candidate group	Hardware	Primary skill outcome
0	All candidates	Simulator, VR or screen, foot/EMG/gaze input	Command mapping, extra-limb attention, timing, task grammar
1	Early learners	Extra thumb or extra finger, low force	Foot-hand dissociation, additional effector ownership, grip sequencing
2	Older trainees	Passive or quasi-passive short module	Spatial awareness, stow/deploy habits, safe body envelope
3	Advanced adolescent or adult trainee	Low-force single TALI-H or TALI-W training module	One-module coordination with biological arms
4	Adult candidate	Adult Alpha single-module wearable	Functional hold, brace, grasp, tool support
5	Qualified adult candidate	Dual module TALI-H/TALI-W or dual TALI-H/TALI-W pair	Quadrimanual coordination and task-specific specialisation

Table 11. Candidate development pathway.

The training system is a project subsystem, not a later educational add-on. TALI-Train should store each candidate profile, calibration data, input-channel history, preferred mappings, skill progression, fatigue trends, and module-specific performance. Training should teach a stable command grammar across TALI-H and TALI-W so that switching modules changes function but not the entire human-machine language.

First Adult Working Prototype

The first adult working prototype is designated TALI-Alpha-1. Its purpose is to prove the integrated platform in competent adult human operation. It is not a paediatric device, commercial industrial product, clinical device, or final morphology. It is the first working adult exosuit demonstrating the shared core, dock interface, one attachable module, hybrid power, body-signal control, mechanical rest/lock, and modular end-effector concept.

TALI-Alpha-1 Required Configuration

Subsystem	Alpha-1 requirement
Core frame	Thoraco-pelvic frame with posterior spine shell, iliac belt, sternum stabiliser, pressure-sensing contact pads, and emergency doffing.
Dock	At least one functional Universal Thoracic Dock; second dock may be mechanical dummy or passive test dock.
First active module	One TALI-H support-arm module or one TALI-W short wing-arm module; TALI-H gripper-first configuration is recommended for earliest adult success.
Degrees of freedom	Three powered gross-positioning DOF plus powered end-effector; lockable passive wrist or terminal node.
Power	External 48 V-class power supply or equivalent; small onboard battery for compute, sensors, controller, and safe retract.
Control	sEMG, foot/toe input, IMU posture tracking, head/gaze target selection, manual mode switch, and E-stop.
End-effector	Rugged two-finger gripper or brace pad first; adaptive hand as second end-effector.
Software	TALI-OS with module ID, candidate profile, joint limits, force/speed limits, rest/deploy/hold/release primitives, and logging.
Operating envelope	Slow, force-limited, adult-supervised engineering checkout. No autonomous high-force motion near head, neck, or bystanders.

Subsystem	Alpha-1 requirement
Deliverable proof	Candidate wears frame, deploys module, performs hold/brace/grasp task, returns to rest, swaps or demonstrates swap-ready dock protocol.

Table 12. TALI-Alpha-1 required configuration.

Alpha-1 Performance Requirements

Requirement class	Minimum adult prototype target
Don/doff	Single trained adult can don with assistant and remove rapidly using emergency release.
Static comfort	No local hard pressure, rib restriction, axillary loading, or neck loading during static wear.
Module deployment	Module deploys from rest to safe ready posture without entering face/neck exclusion zone.
Resting energy	Rest posture uses mechanical lock or brake rather than continuous high motor torque.
Command success	Candidate can command rest, deploy, hold/brace, release, and retract through body-signal interface.
Grasp/brace task	Module holds, clamps, braces, or supports a test object while biological hands remain free.
Logging	System records module state, input channels, joint states, power state, pressure zones, and faults.
Swap readiness	Dock can mechanically and electronically identify at least two module identities or one module plus dummy module.
Adult operation boundary	Prototype use remains limited to competent adults until post-prototype controls and development pathways are completed.

Table 13. TALI-Alpha-1 minimum performance requirements.

Component Candidate Matrix

The following components are candidates, not mandatory final suppliers. They demonstrate that the project can be grounded in existing industrial and research hardware while maintaining a custom exoframe and dock architecture. Selection should be based on mass, payload, protocol access, durability, serviceability, software integration, and fit within the UTD mechanical envelope.

Subsystem	Candidate class or example	Reason for inclusion
Adaptive hand	qb SoftHand Industry	Industrial anthropomorphic hand; tendon-driven; 19 DOF through one synergy; 0.99 kg; IP65; ROS/ROS2 compatibility (qbrobotics, n.d.).
Rugged gripper	Robotiq 2F-85	85 mm stroke, 5 kg payload, 20-235 N grip force, approximately 1 kg mass, 24 V operation (Robotiq, n.d.).
Quick changer	OnRobot Quick Changer or equivalent	200 g class, ISO-9409-1 flange, redundant locking, up to 20 kg payload (OnRobot, n.d.).
sEMG	Delsys Trigno or equivalent	Wireless EMG platform for real-world capture, robotics, human-machine interaction, and API-enabled integration (Delsys, n.d.).
Dry EEG	Wearable Sensing DSI-24	Wireless dry EEG, rapid setup, full head coverage, auxiliary inputs, and API support (Wearable Sensing, n.d.).
Open neurointerface	OpenBCI Cyton or equivalent	Lower-cost EEG/EMG/EKG acquisition ecosystem for exploratory development and education (OpenBCI, n.d.).
Middleware	ROS 2 plus ros2_control	Modular robotics framework with hardware interfaces, transforms, actions, testing, and controller architecture (Open Robotics, 2026; ROS Controls, 2026).

Subsystem	Candidate class or example	Reason for inclusion
Safety standard basis	ISO 12100, ISO 13482, ISO/TS 15066, ISO 10218 family	Design references for machinery risk reduction, personal care robots, collaborative contact, and industrial robot safety (International Organization for Standardization, 2010, 2014, 2016, 2025).

Table 14. Candidate components and enabling standards.

Manufacturing, Materials, and Configuration Management

The TALI system should be manufactured as a modular research-grade engineering platform with controlled configuration, not as a single handcrafted prototype. Each major subsystem should have a part number, revision, serial number, mass budget, calibration record, firmware version, and service interval. The frame, docks, modules, end-effectors, sensors, and software profiles must be traceable because candidate training and module swapping depend on repeatability.

Material Strategy

Part class	Preferred materials	Design reason
Posterior spine shell	Carbon-fibre composite, glass-fibre composite, or aluminium-composite hybrid	High stiffness-to-mass and contoured load transfer
Pelvic belt structure	Machined aluminium, carbon composite, or reinforced polymer with metal inserts	High load transfer and repeatable adjustment
Dock structural core	Machined aluminium or titanium alloy with replaceable wear interfaces	High precision, fatigue resistance, and serviceability
TALI-H links	Carbon tubes, aluminium links, protective covers	Low inertia and sufficient stiffness
TALI-W spars	Carbon-fibre tubes, composite shells, thin aluminium truss elements	Long reach with low distal mass
Pads and garment	Breathable textile, spacer mesh, closed-cell foam, washable liners	Comfort, hygiene, and pressure distribution
Cable routing	Low-friction Bowden sheaths, aramid or steel cables, serviceable pulleys	Remote actuation and low distal mass
Covers	Black-and-white or neutral polymer covers, no sharp exposed mechanisms	Contact protection and submission-ready appearance

Table 15. Material strategy.

Configuration Control

Every module shall declare its module class, side, serial number, software driver, maximum allowed torque, workspace profile, end-effector family, mass, centre-of-mass estimate, calibration date, and fault status before activation. The control system shall not energize an unrecognized module. Adult prototype operation may include engineering override modes, but those modes shall be logged and physically keyed to prevent accidental use during training.

Post-Adult-Prototype Protective Expansion Controls

The project instruction is that broad protective gating and limitation analysis follow the first working adult prototype rather than preventing that prototype from being built. Therefore, this section is positioned after the

adult prototype definition. The official position is that TALI-Alpha-1 is built with intrinsic engineering safeguards, but expansion beyond adult controlled operation requires protective controls, certification analysis, and domain-specific governance.

After the first adult prototype functions, the project should apply staged controls for higher loads, dual modules, industrial environments, paediatric developmental programs, public demonstrations, clinical-adjacent settings, and any aerospace or tethered aerodynamic work. ISO 12100 provides a general machinery risk assessment and risk reduction methodology (International Organization for Standardization, 2010). ISO 13482 covers physical-assistant and personal care robot hazards and human-contact applications (International Organization for Standardization, 2014). ISO/TS 15066 addresses collaborative robot contact applications, and ISO 10218-1:2025 covers industrial robot safety requirements (International Organization for Standardization, 2016, 2025). These standards should not be treated as reasons to abandon the first prototype; they should be treated as the downstream language for scaling it responsibly.

Expansion domain	Protective control after Alpha-1
Dual active modules	Independent module load monitoring, mutual collision avoidance, body-zone exclusion, expanded E-stop logic.
Heavy equipment	External structural bracing, certified grippers, task hazard analysis, operator training.
Industrial workplace	Industrial robot safety assessment, lockout/tagout interface, collaborative force limits, environmental ingress protection.
Surgery-adjacent support	Sterilisable distal tools, sterile drape strategy, low-latency teleoperation, clinical device classification review.
Paediatric/developmental training	Low-force curriculum devices, growth-adjustable frames, age-specific pressure and load limits, guardian and institutional approval.
Public demonstration	Physical barriers, reduced force mode, scripted trajectories, no bystander contact.
Aerospace/tethered aerodynamic work	Separate aviation/structural protocol, no assumption of untethered human flight.

Table 16. Post-Alpha-1 protective expansion controls.

Implementation Master Plan

The master plan is organised around three simultaneous engineering tracks: TALI-Core, TALI-H, and TALI-W. TALI-Core is the shared platform and therefore has priority. TALI-H is the first manipulation model and should deliver the earliest adult functional success. TALI-W proceeds as a separate model with passive and bench prototypes before high-load wearable operation. TALI-Train supports both.

Track C: TALI-Core

- Define adult anthropometric envelope, frame contact zones, and thoraco-pelvic load path.
- Build soft garment and pressure-mapped fitting system.
- Build posterior spine shell and pelvic belt with mechanical dummy dock loads.
- Develop left/right Universal Thoracic Dock mechanical standard.
- Integrate power, data, safety, and module ID bus.
- Commission TALI-OS, candidate profiles, and simulator link.

- Validate adult don/doff, pressure, comfort, and dummy-load carriage.

Track H: TALI-H

- Bench-test rugged gripper and adaptive hand candidates.
- Build fixed-base single TALI-H arm with three powered gross DOF and quick-change endpoint.
- Integrate sEMG/foot/gaze shared-control primitives.
- Move heavy actuation to torso/root location and minimise distal mass.
- Mount one TALI-H module to TALI-Core through UTD.
- Demonstrate adult hold, clamp, brace, release, and rest operations.
- Add second asymmetric module after single-module adult success.

Track W: TALI-W

- Build passive foldable wing-arm dummy to test stowage, body clearance, and dock loads.
- Build bench-mounted powered root and fold joint with carbon spar geometry.
- Test brace endpoints, hooks, cradles, and jamming or clamp endpoints.
- Mount short low-force TALI-W module to TALI-Core for adult fit and deployment.
- Demonstrate fold, deploy, brace, stabilise, retract, and rest modes.
- Scale spar length and load only after root, dock, and pelvic load transfer are proven.
- Keep tethered aerodynamic experiments as a separate downstream track.

Track T: TALI-Train

- Create simulator with virtual TALI-H and TALI-W modules.
- Create common command grammar and module-specific behavior libraries.
- Introduce foot, sEMG, gaze, IMU, and optional EEG input modes.
- Build extra-thumb or extra-finger starter devices for early motor skill training.
- Record candidate learning, fatigue, error, control stability, and preferred mappings.
- Use curriculum data to generate adult module calibration profiles.
- Support longitudinal candidate development without forcing early users into high-load suits.

Milestone Table

Milestone	Deliverable	Completion criterion
M0	Project architecture freeze	TALI-Core, TALI-H, TALI-W, UTD, and TALI-Train definitions approved.
M1	Simulator and control rig	Candidate can command virtual rest, deploy, hold/brace, release, and retract.

Milestone	Deliverable	Completion criterion
M2	Core frame dummy-load prototype	Adult can wear frame with dummy dock loads without forbidden load paths.
M3	Bench TALI-H module	Module manipulates or grips objects under shared control on fixed stand.
M4	Bench TALI-W module	Wing-arm folds, deploys, braces, and retracts on fixed stand.
M5	UTD integration	Core detects module ID and loads correct profile before energizing.
M6	Adult single-module wearable	Adult candidate completes wearable hold/brace/grip task and returns to rest.
M7	Module interchange demonstration	TALI-H, TALI-W, or dummy module can be changed through non-loaded dock protocol.
M8	Dual-module adult platform	Two modules operate with mutual collision limits and shared command grammar.
M9	Post-prototype expansion package	Protective controls, standards mapping, domain-specific pathway, and training certification prepared.

Table 17. Master implementation milestones.

Official First-Build Recommendation

The official first build should be TALI-Alpha-1H: TALI-Core with one active TALI-H support module, one UTD, external power, a rugged gripper, body-signal control, mechanical rest locking, and software logging. This is the highest-probability adult prototype because it demonstrates practical value quickly: the module can hold, clamp, stabilise, or support equipment while the candidate uses both biological hands. Once TALI-Alpha-1H is successful, the same TALI-Core can accept a passive or low-force TALI-W module to demonstrate the separate wing-arm morphology without mixing the two models.

The second build should be TALI-Alpha-1W: TALI-Core with one active short TALI-W module. Its adult demonstration should be fold, deploy, brace, stabilise, and retract. Only after both single-module adult prototypes function should the project build dual asymmetric operation. The first dual system may combine one TALI-H and one TALI-W, but the modules remain distinct and are managed by the UTD and TALI-OS as separate model classes.

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Appendix A: Requirements Register

ID	Requirement	Applies to	Priority
R-001	The system shall use a thoraco-pelvic frame rather than a rib-only frame.	Core	Critical
R-002	The system shall keep TALI-H and TALI-W as separate module classes.	All	Critical
R-003	The system shall provide at least one Universal Thoracic Dock for Alpha-1.	Core	Critical
R-004	The dock shall identify attached module class before energizing the module.	Dock/Software	Critical
R-005	The frame shall transfer primary loads through posterior spine shell and pelvic belt.	Core	Critical
R-006	The first adult prototype shall use external energy for high-load operation.	Power	Critical
R-007	Resting states shall use mechanical lock, brake, or passive support rather than high continuous motor current.	Power/Module	Critical
R-008	The system shall include non-surgical body-signal control channels.	Control	Critical
R-009	EEG shall be supervisory, optional, and not the sole low-level controller in Alpha-1.	Control	High
R-010	TALI-H shall support at least one rugged gripper endpoint.	TALI-H	Critical
R-011	TALI-W shall support fold, deploy, brace, stabilise, retract, and rest.	TALI-W	Critical
R-012	Modules shall not be hot-swapped while loaded or energized.	Dock	Critical
R-013	The software shall load module-specific kinematics, limits, and command mappings.	Software	Critical
R-014	The garment shall support repeatable sensor placement and pressure telemetry.	Human interface	High
R-015	The system shall preserve natural biological arm movement and scapular clearance.	Core	High
R-016	The system shall include emergency stop and emergency doffing.	All	Critical
R-017	The training environment shall support virtual TALI-H and TALI-W modules.	Training	High
R-018	Developmental training shall begin with simulation and low-force devices, not high-force child exosuits.	Training	High
R-019	All modules shall have serial-numbered configuration records.	Configuration	High
R-020	The first adult prototype shall demonstrate human-worn rest, deploy, hold/brace, release, and retract.	Alpha-1	Critical

Table A1. Initial requirements register.

Appendix B: Interface Control Summary

Interface	Mandatory signals or features
UTD mechanical	Keyed geometry, redundant latch, load rating, release status, module side index.
UTD electrical	Power enable, pre-charge, ground, protected supply rails, module fault line.
UTD data	Real-time bus, module ID, firmware version, calibration hash, heartbeat.
UTD fluid	Optional vacuum/pneumatic/cooling line with capped safe default.
Candidate garment	sEMG mounts, IMU pockets, pressure pads, cable channels, size markers.
End-effector node	Quick-change interface, tool ID, force sensor channel, safe release state.
Training system	Candidate ID, module class, input mappings, performance metrics, log export.

Table B1. Interface control summary.

Appendix C: Adult Alpha Demonstration Task Set

Task	Description	Pass criterion
Wear and fit	Adult candidate dons TALI-Core with assistant.	No forbidden pressure zone, frame stable, sensors online.
Rest posture	Module remains folded and locked.	No continuous high-current holding required.
Deploy	Candidate commands deploy to ready posture.	Module reaches posture without entering exclusion zone.
Hold object	TALI-H grips or TALI-W braces a standard test object.	Object held or surface braced for defined interval.
Biological dual task	Candidate uses both natural hands while module maintains hold or brace.	Module remains stable and candidate can complete task.
Release	Candidate releases or unloads endpoint.	No uncontrolled motion or dropped hazardous load.
Retract	Module returns to rest.	Rest lock engaged and software state updated.
Fault response	Signal loss or E-stop is triggered under unloaded test.	Module enters designed fault state and logs event.

Table C1. Adult Alpha demonstration task set.

Appendix D: Model Separation Rules

- TALI-H and TALI-W shall use separate CAD assemblies, kinematic definitions, module IDs, and software plugins.
- No TALI-W spar shall be described as a hand-arm unless fitted with a tool endpoint that performs a hand-arm task.
- No TALI-H arm shall be described as a wing merely because it folds or stows.
- Mixed operation is allowed only at the system level: one TALI-H module and one TALI-W module may be attached to TALI-Core as distinct modules.
- Training commands may be shared, but module behavior must remain class-specific.
- Component selection may overlap only when a component is genuinely generic, such as a quick changer, sensor, controller, or actuator class.
- Drawings, documentation, and test logs shall label each module by class and serial number.

Appendix E: Submission Statement

This document presents the TALI Modular Thoracic Auxiliary Limb Exosuit as an official design master plan. It defines the project goals, product family, shared platform, two distinct interchangeable models, adult prototype path, hardware architecture, control architecture, training curriculum, and implementation milestones. It intentionally preserves the full project information supplied for the final research design and does not collapse the work into a summary. The paper is prepared in black-and-white APA-style formatting with hyperlinked references for submission use.